Putting Patterns, Data and Points Together

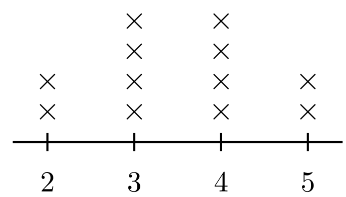
Pattern rules, reading the plot, coordinate pairs, and summarising data

Directions

- Problems 1 to 4 warm up on the two pattern rules. After that the four kinds are mixed — decide which display and which method each problem needs before you start.
- Show the terms you generated or the counts you read before writing an answer.

Term	1	2	3	4	5	6
S	3	7	11	15	19	23
T	1	7	13	19	25	31

Display 1 — Two Rules. Pattern S: start at 3, add 4. Pattern T: start at 1, add 6.



Display 2 — Book Plot. Books read last week by each of 12 students. One × is one student.

1. Use Display 1. Write the 8th term of pattern S.

Answer: ________________

2. Use Display 1. Write the 8th term of pattern T.

Answer: ________________

3. Use Display 1. At which term number do patterns S and T give the same value?

Answer: ________________

4. Use Display 1. How much greater is pattern T's 10th term than pattern S's 10th term?

Answer: ________________

5. Use Display 2. How many students read exactly 3 books?

Answer: ________________

6. Use Display 1. Write the ordered pair (pattern S's 5th term, pattern T's 5th term).

(________ , ________)

7. Use Display 2. How many books did the 12 students read altogether?

Answer: ________________

8. Use Display 2. How many students read more than 3 books?

Answer: ________________

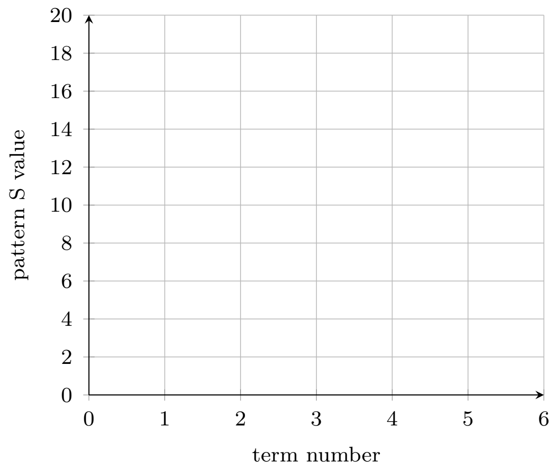
9. Use Display 2. How many more students read 3 books than read 5 books?

Answer: ____________________

10. Use Display 1 and the grid below.

(a) Write the y-coordinate of the ordered pair for term 4 of pattern S: (4, __________)

(b) Plot the ordered pairs (term number, pattern S value) for terms 1 to 4 on the grid.



11. Use Display 2.

(a) Write the median number of books read.

Answer: ____________________

(b) Write the mean number of books read.

Answer: ____________________

12. A reading club's weekly totals follow pattern S from Display 1, counted in books per week.

(a) How many books did the club read in week 6?

Answer: ________________

(b) How many books did the club read over the first 6 weeks altogether?

Answer: ________________

(c) Explain how the mean of the six weekly totals could be used to check your answer to part (b).

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